Cloud Security Project Report: AWS EC2 Security Hardening & Monitoring Lab

Author: Nitin Sukthe

Project Type: Cloud Security / AWS / Linux Hardening

Executive Summary

This project demonstrates secure deployment, hardening, and monitoring of an AWS EC2 instance using Amazon Linux 2023. Security controls implemented include restrictive Security Groups, SSH key authentication, Linux hardening, least■privilege access control, and AWS CloudWatch monitoring.

Project Objectives

• Deploy a secure AWS EC2 environment
• Configure secure network access controls
• Implement SSH key authentication
• Harden Linux configuration
• Configure monitoring and alerting

Environment & Technologies

AWS EC2, Amazon Linux 2023, Security Groups, SSH, Linux Administration, AWS CloudWatch, SNS Notifications.

Implementation Methodology

1. EC2 deployment in ap-south-1 (Mumbai).
2. Security Group configured with SSH access restricted to My IP.
3. SSH connection established using private key authentication.
4. Linux server updated and hardened.
5. Dedicated admin user (cloudadmin) created and assigned wheel privileges.
6. SSH configuration hardened: root login disabled, password auth disabled, MaxAuthTries limited.
7. CloudWatch CPU alarm configured with SNS email notifications.

Security Controls Implemented

Identity & Access: SSH key authentication, dedicated admin user, least privilege.
Network Security: Security Group restrictions, SSH trusted IP allow■listing.
Host Security: Package updates, SSH hardening, root login disabled.
Monitoring: CloudWatch metrics, CPU alarms, SNS notifications.

Key Commands

sudo dnf update -y
sudo adduser cloudadmin
sudo usermod -aG wheel cloudadmin
sudo nano /etc/ssh/sshd_config
sudo systemctl restart sshd

Risk Analysis

Potential risks addressed include unauthorized SSH access, brute■force attacks, weak authentication practices, and lack of monitoring visibility.

Lessons Learned

Hands■on experience gained in AWS EC2 deployment, Linux hardening, cloud monitoring, security configuration, and operational security best practices.

Future Improvements

IAM Roles, CloudTrail Logging, AWS GuardDuty, AWS Config Rules, Terraform automation, vulnerability scanning.

Conclusion

This project demonstrates foundational cloud security engineering skills aligned with entry■level cloud security and SOC/cloud operations roles.